Transporting quadratic products across central charge windows Output-deficit norms and a coherent-cross calculation

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Abstract

An adjacent-charge Hilbert–Schmidt isometry transfers the normalized quadratic product from half filling to a compact central window. The coherent-force norm bounds each step; a sixth output moment controls the accumulated deficit-weighted error. This proves the central-window Gram on the energy square, input deficit and output deficit, with an explicit $O(p^{-2/3})$ transport bound for the last field. A candidate remaining force cross is supported by exact finite CAR contractions and full-rank sums. Its general-charge Casimir recoupling step is isolated at the end for further proof.

1 State, normalization and the established result

Use the full overlapping complex Gaussian p-body Hamiltonian, standard CAR, $\hbar = 1$, and normalized maximally mixed trace in charge m. Norms are annealed: $\|T\|_{2,m}^2 = \mathbb{E}\,\mathrm{tr}_m(T^\dagger T)$. All sectors share the same Gaussian couplings and variance w_N. Take even N and

$$\lambda_N = p^2/N \to \lambda > 0, \quad m = N/2 + xp, \quad |x| \le L, \qquad q = e^{-4\lambda}. \tag{1}$$

Bounds are uniform for λ_N in a positive compact set and integer m in this window. Let h_a be its rank-a Gaussian part, $C_m = \mathbb{E}h^2 = w_N\binom{m}{p}\binom{N-m+p}{p}$, and W_2 denote ordinary Gaussian second chaos. Put

$$B_m = \mathsf{W}_2 h^2, \qquad D_m = \sum_a (p-a)h_a, \tag{2}$$

$$L_{\mathrm{in,m}} = \mathsf{W}_2\{h, D_m - \bar{u}_m h\}, \qquad \bar{u}_m = \sum_a (p-a)\mathbb{E}h_a^2/C_m, \tag{3}$$

$$\mathcal{S}_m = \sum_{a\le b,r} (a+b-r)\mathsf{W}_2 Z^m_{ab;r}. \tag{4}$$

Here $Z^m_{ab;r} = P_r(h_a \odot h_b)$, where $\odot$ is a square for $a = b$ and an anticommutator otherwise. We prove

$$\frac{G(B, L_{\mathrm{in}}, \mathcal{S})}{C_m^2} \longrightarrow \begin{pmatrix} 1+q & 0 & b \\ 0 & 2\lambda(1+q) & 0 \\ b & 0 & s \end{pmatrix}, \quad \begin{cases} b = 2\lambda(1-q), \\ s = 4\lambda^2(1+q) + 2\lambda(1-q). \end{cases} \tag{5}$$

The known half-filled Gram is the reference, not a parity identity assumed at the destination charge.

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2 Exact adjacent-charge transport

Set $n = N - m + 1$ and $x_r = r(N + 1 - r)$. The established removal norm is

$$\| \operatorname{Rem} T_r \|_{2,m}^2 = \ell_{m,r}^2 \| T_r \|_{2,m-1}^2, \qquad \ell_{m,r}^2 = \frac{m}{n}(mn - x_r), \quad \operatorname{Rem} T = \sum_i c_i^\dagger T c_i. \tag{6}$$

Define $\mathcal{I}_m = \sum_{r \le 2p} \ell_{m,r}^{-1} \operatorname{Rem} P_r$. For large p the entire central path satisfies $2p \le \min(m-1, N-m)$, so both sectors contain every rank used here. Equivariance and multiplicity-free neutral operator spaces make $\mathcal{I}_m$ an onto isometry on this common subspace. Its inverse is its normalized-HS adjoint. No assertion about positivity of this map is required.

The map acts only on the output matrix of a Gaussian coefficient tensor. The common p-body coupling space decomposes into the same input ranks in all sectors. Thus input-pair blocks and output ranks are preserved, including their internal multiplicities. The map commutes with W_2 and every block weight $f(a, b, r)$.

Write $\widehat{B}^m_{ab;r} = \mathsf{W}_2 Z^m_{ab;r}/C_m$ and

$$R_m = C_{m-1}/C_m = \frac{(m-p)(n+p)}{mn}, \qquad t_{m,r} = R_m \ell_{m,r}, \tag{7}$$

$$A_m = \operatorname{Rem}(H_{m-1}^2) + (2p - m) H_m^2, \qquad O_m = \mathsf{W}_2 A_m. \tag{8}$$

The same-J product identity is exactly

$$t_{m,r} \mathcal{I}_m \widehat{B}^{m-1}_{ab;r} - (m - 2p) \widehat{B}^m_{ab;r} = O^m_{ab;r}/C_m. \tag{9}$$

In the stated window, uniformly for $r \le 2p$,

$$t_{m,r} \ge cN, \qquad |t_{m,r} - (m - r)| \le K, \qquad |t_{m,r} - (m - 2p)| \le K(1 + u + v + S), \tag{10}$$

where $u = p - a, v = p - b, S = a + b - r$. These follow by squaring t: all scaled denominators are bounded away from zero, and $t^2 - (m - r)^2 = O(N)$. More explicitly, for $r = \rho p$,

$$t_{m,r} - (m - r) \longrightarrow 2\lambda[x(2 - \rho) - 1], p^2(R_m - 1) \longrightarrow 4\lambda^2(2x - 1). \tag{11}$$

The first limit is a scalar check for bounded $\rho \in [0, 2]$; the proof below uses the uniform inequality (10).

3 Telescoping and the output tail

Two inputs from the preceding central-window notes are used:

$$\sum_{a\le b,r}(1+u+v)^2\|O^m_{ab;r}/C_m\|_2^2\le K, \tag{12}$$

$$\sum_{a\le b,r}S^6\|\widehat{B}^m_{ab;r}\|_2^2\le K. \tag{13}$$

There are also uniform factorial tails in u,v. For clarity, (13) follows from the positive ordered-product law π; the symmetrized same-J block is bounded by twice the ordered law after the ordered-pair convention is included. It does not require a sign formula for the exchanged contraction. The same bound holds with each fixed polynomial in $u+v$ inserted, by the conditional uniform sixth moment and the factorial input tail.

Solving (9), and summing squared orthogonal block norms, now gives

$$\|\mathcal{I}_m(B_{m-1}/C_{m-1})-B_m/C_m\|_2\le K/N. \tag{14}$$

Indeed the numerator is $O_m/C_m+(m-2p-t_{m,r})\widehat{B}^m$ in each block; (12), (10) and the product second moment bound it uniformly. The same argument with weight $u+v$ uses (12) and the weighted product moments. Composing the isometries (their inverses below half filling) over at most Lp steps proves an unweighted error $Kp/N=O(p^{-1})$.

Let $U_{m\leftarrow 0}$ denote this composition from $m_0=N/2$. All steps preserve a,b,r, so the cutoff $P_{S\le R}$ is the same coefficient-space projector throughout. Telescope first and then multiply by S on this low block. For the two endpoint tails, (13) gives

$$\|S\mathbf{1}_{S>R}\widehat{B}^m\|_2^2\le K/R^4. \tag{15}$$

Consequently

$$\left\|U_{m\leftarrow 0}\frac{\mathcal{S}_{m_0}}{C_{m_0}}-\frac{\mathcal{S}_m}{C_m}\right\|_2\le K\left(\frac{R}{p}+\frac{1}{R^2}\right)=O(p^{-2/3})\quad(R=p^{1/3}). \tag{16}$$

Only the two endpoint tails are counted; summing a tail bound at every intermediate charge would give a different, inadequate estimate.

The input-weighted version of (14), together with $\bar{u}_m\to\lambda$ uniformly, also transports L_{in}/C_m with an error tending to zero. Thus all three fields in (5) converge in the same coefficient Hilbert space to their half-filled references. Uniform boundedness and Cauchy–Schwarz transfer every Gram entry, proving (5). The displayed rate concerns transport only, not the rate of the reference Gram's DS limit.

4 A candidate remaining coherent cross

The preceding proof establishes the output self norm. To add O_m to the Gram, consider $X = x_{a+b} - x_r$. Since

$$X = S[N + 1 - 2(a + b) + S], \qquad \|\mathcal{X}_m/N - \mathcal{S}_m\|_2/C_m = O(p^{-1}), \tag{17}$$

the uniform O_m/C_m bound permits replacing the deficit by this Casimir proxy in a cross. Define the neutral Racah kernel χ with lower parameters $(-m, m - N, -N - 1)$, and

$$\xi = {}_4F_3\left[\begin{matrix}-a, a - N - 1, -b, b - N - 1 \\ 1 - m, m - N, -N - 1\end{matrix}; 1\right], \qquad D = \xi - \chi. \tag{18}$$

Both kernels have the same displayed upper parameters. Put $g = (m - p)^2/m$, $\eta = p^2/m$ and

$$C_f = \frac{m(N - m)(N + 1)}{N}, \quad C_t = \frac{(m - 1)(N - m)(N + 1)}{N - 1}, \quad U_{ab} = -2ab(1+\chi)+x_a x_b/C_f. \tag{19}$$

The neutral proxy cross per ordered pair is U_{ab}/N. For the coherent product with $\mathrm{Rem}(H^2_{m-1})$, the candidate direct and exchanged Casimir moments, divided by C_aC_b, are

$$g(x_a + x_b - x_a x_b/C_t), \qquad g[(x_a + x_b)\xi - mD]. \tag{20}$$

They pass the independent pure-rank CAR checks described on page 6. If (20) holds generally, algebra gives the stable same-J formula

$$z_{ab} = \frac{1}{N}\left\{\eta U_{ab} + g\left[\frac{x_a x_b}{m(m - 1)(N + 1)} + (m - 2ab)D\right]\right\}. \tag{21}$$

It combines the large terms before evaluation, using $C_t^{-1} - C_f^{-1} = [m(m - 1)(N + 1)]^{-1}$.

For fixed input deficits, terminating-series bounds give $\chi \to q$ and $mD \to -4\lambda q$. Each term is bounded by $K^j/j!$, including one extra j moment. Thus (21) is uniformly bounded over all $a, b \le p$ and tends to s. The factorial input tail justifies the full rank sum. Combined with (17), this would prove $\langle \mathcal{S}_m, O_m\rangle/C_m^2 \to s$. This last inference is conditional on the general-charge recoupling identity (20), not on numerical convergence.

5 The precise analytic step still needed

The charged covariance is diagonal with terminating kernel

$$T_a(t) = {}_4F_3\left[\begin{matrix}-a, a-N-1, -t, t-N \\ 1-m, m-N, -N-1\end{matrix}; 1\right]. \tag{22}$$

The established two-component decomposition gives

$$\xi = T_a(b) - \frac{b}{N+1-2b}[T_a(b-1) - T_a(b)], \tag{23}$$

$$\frac{x_b}{N+1-2b}[T_a(b-1) - T_a(b)] = -mD. \tag{24}$$

The second equality follows termwise: if c_j is a summand of χ, the corresponding summand of ξ is $mc_j/(m-j)$. We have additionally checked 624 finite rational instances.

What remains is to identify the exchanged gradient contraction with the left side of (24) times g. With $E_{ij} = c_i^\dagger c_j$ and $\mathcal{C} = \frac{1}{2}\sum_{ij} \mathrm{ad}_{E_{ij}} \mathrm{ad}_{E_{ji}}$, the exact product rule is

$$\mathcal{C}(UV) = (x_a + x_b)UV + \sum_{ij}[E_{ij}, U][E_{ji}, V]. \tag{25}$$

The needed evaluation uses the two charged components of Vc_i and the CAR commutators $[E_{ij}, c_k] = -\delta_{ik}c_j$. It must track the rectangular trace normalization and both component weights. The termwise identity alone does not evaluate this operator contraction. The half-filled alternating-completeness proof in the earlier note does not supply that general-charge step.

For a concrete continuation, set $z = 1+q$, $l = 2\lambda(1+q)$, $a_* = 4\lambda + 4\lambda^2(1+q)$. The established subblocks (5) and the preceding force note leave exactly one unknown:

$$C_m^{-2} G(B, L_{\mathrm{in}}, \mathcal{S}, O) \longrightarrow \begin{pmatrix} z & 0 & b & b \\ 0 & l & 0 & 0 \\ b & 0 & s & c \\ b & 0 & c & a_* \end{pmatrix}. \tag{26}$$

Existence and value of c are the remaining cross problem. The candidate is $c = s$. Under that condition, writing

$$v = s - b^2/z = 2\lambda(1-q) + \frac{16\lambda^2 q}{1+q}, \qquad \Delta = a_* - s = 2\lambda(1+q), \tag{27}$$

the residual of $\mathcal{S}/C_m$ after projection onto $\mathrm{span}(B, L_{\mathrm{in}}, O)$ has variance $v\Delta/(v+\Delta) > 0$. This is a conditional Schur complement. Its central-window status must remain distinct from the already established half-filled residual.

6 Reproducible checks and continuation

The new files in `results/susskind-memory-central-cross` record the following independent scopes:

- `transport.json`: exact symbolic scalar drift and variance ratio; complete rational isometry matrices on ranks $0, 1, 2$ for $(N, p, m) = (6, 1, 3)$, dimensions $1, 35, 189$. Full 1,296 ordered Gaussian coordinate pairs verify the normalized transport identity with maximal error $3.47\,10^{-17}$; the square-root isometry norm comparison differs by $2.23\,10^{-16}$.
- `charged-casimir.json`: 15 pure-rank pairs in four small systems, all 60 direct/exchanged zeroth/first Casimir identities exact. Across these blocks 66,531 ordered Gaussian coordinate pairs are contracted, without a parity substitution.
- `car-exploration.json`: independent full 4×4 coefficient Grams on $(B, \mathcal{S}, \mathcal{X}/N, O)$, totaling 31,466 symmetric coordinate pairs. All four differences between the actual proxy–force cross and the candidate are exactly zero. Its original exploratory status is retained.
- `cross-limit.json`: 624 rational contiguous checks, four finite all-rank exact comparisons, and 36 central cases at $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, $x = -2, 0, 2$. Every input rank and terminating-series term is retained. Evaluations at 80 and 120 digits agree at the stored 65 significant digits. The three columns are $\mathcal{X}/N$ crossed with B, L_{in} and the candidate O expression, respectively.

At $\lambda = 0.5, p = 128$, the candidate force cross divided by C_m^2 is approximately 1.97682736, 1.94517151, 1.91541182 for $x = -2, 0, 2$; its proposed limit is 2. These finite-size deviations are much larger than precision-refinement differences. Neither floating refinement nor exact small-system checks prove the unresolved recoupling lemma.

The transport theorem uses even N and a fixed compact central window. Odd-N references, growing charge windows and full charge sums require additional work. The next immediate task is the gradient contraction on page 5, followed by energy insertions, the raw memory spectrum and finite-memory remainder. The space/mass dictionary for Susskind's bulk mesons remains downstream.

References

[1] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1, Eqs. (2.11)–(2.19). The irreducible covariance and fixed-charge framework are prior results; particle–hole symmetry does not imply a same-sector product parity away from half filling.

[2] A. Day and Chengheng, local project notes: *Parity-weighted rank deficits*, *Central charge windows*, and *Coherent memory norms on central charge windows* (6 November 2026). These provide the reference Gram, positive sixth moment and coherent-force bound used in the transport proof.